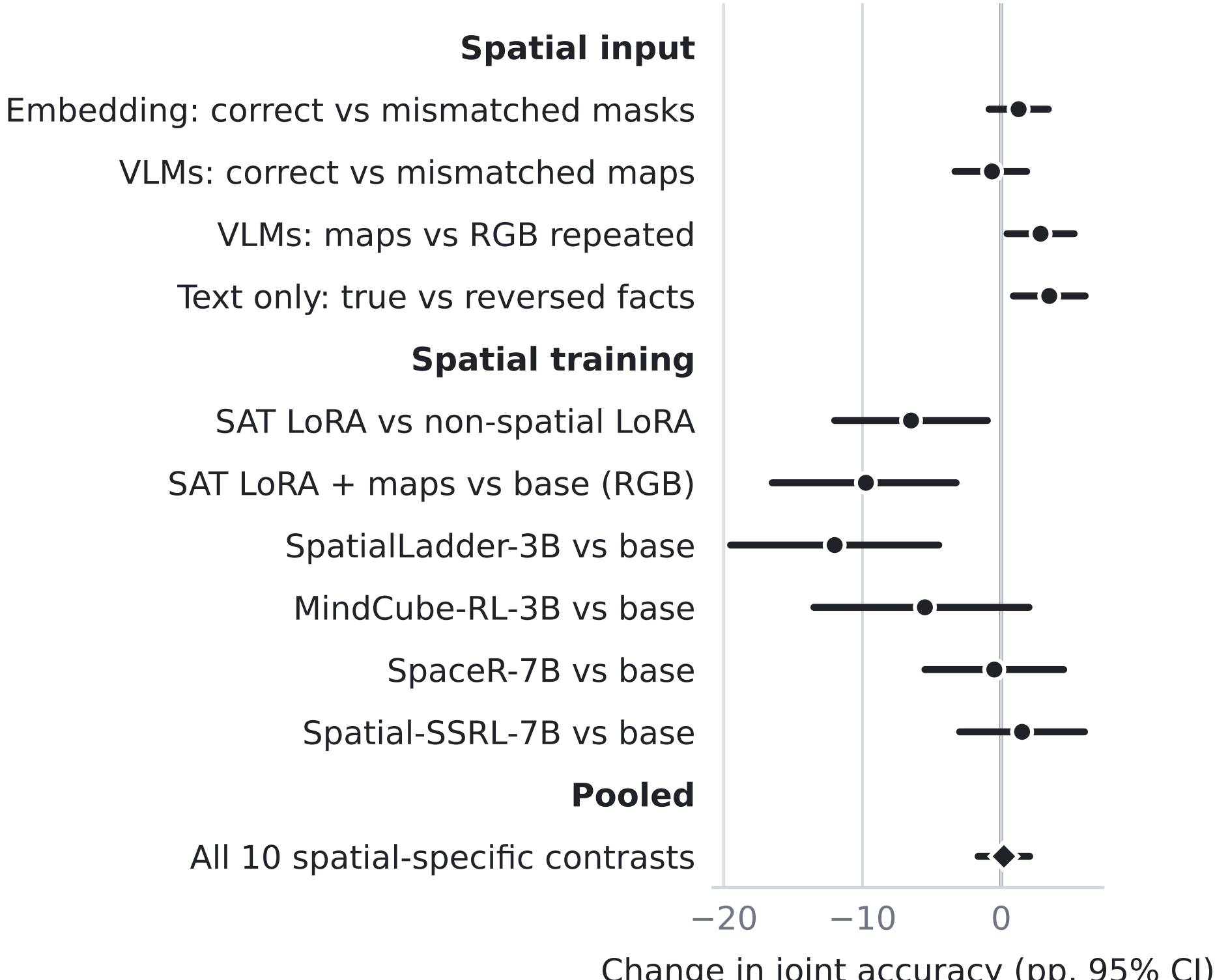
Spatial input
Embedding: correct vs mismatched masks
VLMs: correct vs mismatched maps
VLMs: maps vs RGB repeated
Text only: true vs reversed facts
Spatial training
SAT LoRA vs non-spatial LoRA
SAT LoRA + maps vs base (RGB)
SpatialLadder-3B vs base
MindCube-RL-3B vs base
SpaceR-7B vs base
Spatial-SSRL-7B vs base
Pooled
All 10 spatial-specific contrasts
−20
−10
0
Change in joint accuracy (pp, 95% CI)